

Venkat Aswin E

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PROFESSIONAL SUMMARY

Final-year Computer Science student specialising in AI engineering, with a published conference paper on multimodal disaster classification (92% accuracy across 3 benchmark datasets). Delivered measurable results in RL (40% win-rate improvement), RAG pipelines, and multi-agent LLM systems. Certified in AWS Cloud Architecting, Microsoft Azure AI, and Oracle OCI Generative AI.

EDUCATION

Vellore Institute of Technology — Amaravati <i>B.Tech, Computer Science</i>	Aug. 2022 – May 2026 <i>GPA: 8.03 / 10</i>
FIITJEE Junior College, Hyderabad <i>Class XII — State Board</i>	Jun. 2020 – Apr. 2022 <i>Score: 88%</i>
DAV Public School <i>Class X — CBSE</i>	Apr. 2020 <i>Score: 91%</i>

TECHNICAL SKILLS

Languages: Python, Java, SQL
ML / DL: PyTorch, TensorFlow, Scikit-learn, Stable-Baselines3, Optuna, SHAP
LLM & GenAI: LangChain, Hugging Face Transformers, Ollama, Prompt Engineering, LoRA / QLoRA Fine-Tuning
RAG & Agents: RAG Pipelines, FAISS, ChromaDB, Sentence Transformers, Multi-Agent Orchestration, ReAct Agents
Specialisations: Reinforcement Learning, CNNs, Vision Transformers, Multimodal AI, Graph Neural Networks
Cloud / Tools: AWS, Azure AI, Oracle OCI (GenAI), Flask, Docker, Git

PROJECTS

- AI-Powered Disaster Management & Geospatial Dashboard** | *PyTorch, ViT, CNNs, GeoPandas, Folium, Plotly*
- Achieved **92% classification accuracy** on held-out test data by training a multimodal AI system fusing satellite imagery, captions, and geospatial metadata (NDVI, NDBI) across 3 disaster lifecycle phases.
 - Outperformed single-modality baselines by combining CNNs and Vision Transformers in a hybrid late-fusion architecture augmented with remote sensing spectral indices.
 - Constructed an interactive geospatial dashboard ingesting **3 live hazard feeds** (USGS, GDACS, NOAA) to render real-time confidence maps and early-warning alerts.
 - Increased model transparency for emergency-response stakeholders by applying Grad-CAM, attention heatmaps, and saliency maps; results submitted as a conference paper (2025).
- Street Fighter II Reinforcement Learning Agent** | *Python, Gym-Retro, Stable-Baselines3, Optuna*
- Raised agent win-rate by **40%** and mean episode reward by **27%** through reward shaping, frame-stacking, and state preprocessing over 30k+ training timesteps.
 - Implemented a PPO training pipeline with a 3-layer CNN policy and 512-unit MLP head, achieving stable convergence under sparse and delayed reward signals.
 - Reduced hyperparameter convergence time by **35%** by automating search across 8 parameters with Optuna (50+ trials); identified policy collapse patterns via entropy decay analysis.
- DataMind — Autonomous AI Data Analyst** | *Python, Flask, LangChain, Ollama, Pandas, Scikit-Learn, Plotly.js*
- Designed a **7-agent** ML pipeline — schema understanding, cleaning, EDA, visualisation, modelling, insight generation, and a chat interface — powered by local Ollama LLMs.
 - Improved answer relevance and reduced hallucinations by integrating RAG-style context retrieval grounded in dataset-specific embeddings.
 - Benchmarked Random Forest and Logistic Regression models via cross-validation leaderboards, surfacing the top features automatically through an ML Agent with Pearson correlation screening ($|r| > 0.7$).
 - Shipped a responsive single-page frontend (HTML5/CSS3/JS) with **5+ Plotly.js** chart types and an interactive query panel for on-demand data exploration.

EXPERIENCE

Eminent Software Solutions	Jun. 2024 – Aug. 2024
<i>SAP Development Intern (Remote)</i>	
• Cut average report execution time by 25% by optimising SAP ABAP queries with CDS Views through systematic profiling and index tuning. • Collaborated with cross-functional teams across Sales, Purchasing, and Finance to implement Passive Screen Programming enhancements, reducing user-interaction latency by 15%. • Delivered custom SmartForms templates for 3+ business units, coordinating testing cycles and stakeholder sign-off independently.	

RESEARCH & PUBLICATIONS

Integrating AI and Geospatial Mapping for Rapid Disaster Detection and Assessment	2025
<i>Conference Paper VIT-AP University, School of Computer Science and Engineering</i>	
• Co-authored with 5 researchers; accepted at VIT-AP University SCSE conference covering a Glass-Box AI framework (CNNs + Vision Transformers + spectral indices) with Grad-CAM explainability. • Achieved 92% classification accuracy across 3 benchmark datasets: FloodNet, BigEarthNet, and So2Sat-LCZ42.	

CERTIFICATIONS

AWS Academy Cloud Architecting	2024
Microsoft Azure AI Fundamentals (AI-900)	2024
Oracle OCI Generative AI Professional	2024